



Introduction to rotations

- 1 In which direction will this vehicle be facing when it is rotated so its wheels touch the road? Tick **1** correct answer

☐

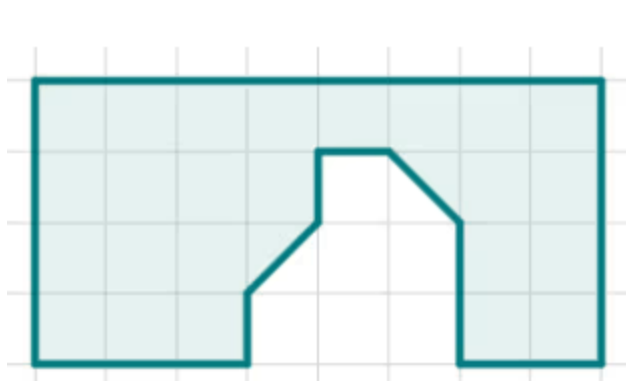
left



right



- 2 Which of these objects, when rotated by a quarter-turn, will fit into the space in this shape to complete a rectangle? Tick **1** correct answer



☐


A

☒


B

☐


C

☐


D

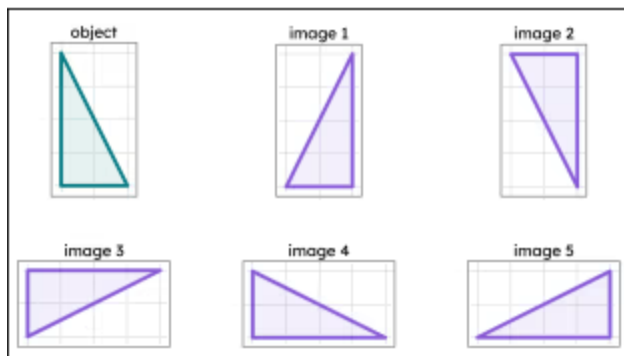
- 3 Complete this sentence: "When an object is rotated, the lengths of its sides are invariant. This means the lengths of the sides remain the same." Fill in the blank

- 4 Sam transforms an object and its orientation changes. Which of these describes why the orientation has changed? Tick **2** correct answers

- ☐ The object hasn't been rotated to become its image.
- ☐ The object has been translated right to become its image.
- ☒ The object has been rotated through a half turn to become its image.
- ☐ Orientation always changes when objects are translated.
- ☒ Orientation always changes when objects are rotated by less than a full turn.



5 Which of these is a rotation of this object? Tick **3** correct answers



- ☐ Image 1
- ☒ Image 2
- ☒ Image 3
- ☐ Image 4
- ☒ Image 5

Explain why the position of an object doesn't always change after a rotation.

